INT333:DEVOPS ADVANCE CONFIGURATION MANAGEMENT

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: Understand the basics of Puppet.

CO2 :: Demonstrate configuration management skills using Puppet.

CO3 :: Learn how to install and configure Nagios for continuous monitoring.

CO4 :: Understand and implement infrastructure automation using Terraform.

CO5 :: Describe the fundamentals of Ansible and its installation process.

CO6 :: Construct Ansible playbooks to automate multiple system administration tasks.

Unit I

Puppet Basics : Configuration Management System: Configuration Management, Pull, Push, Introduction to Puppet: Puppet, Why Puppet, Components of Puppet, Puppet Architecture: Puppet Master: Manifest, Template, Files, Certificate Authority, Puppet Clients: Agent, Facter, Installation of Puppet, Puppet Development in Isolation

Unit II

Advanced Puppet : Puppet Configuration, Managing Packages in Puppet, Puppet Modules, Monitoring Web Servers, Load Balancing Clusters, Scaling Puppet Environment, Connecting Puppet Agent with Puppet Master, Making Configuration Dynamic, Extending Puppet, Puppet Classes and Functions (including Custom Functions), Using Puppet via Command Line, Managing Resources with puppet apply Command, Puppet Manifests

Unit III

Nagios Monitoring : Continuous Monitoring Concepts: Definition, Importance, and Benefits. Introduction to Nagios: Features, Architecture, Plugins, Soft and Hard States, Installation of Nagios: Using Package Managers (apt-get/dpkg, yum/rpm), Installing Prerequisites, Compiling and Installing, Setting up Web Server, Command-Line Interfaces, Nagios Configuration: Monitoring Web Servers, Using Built-in Web Interface (Managing Hosts, Services, Downtimes, Comments, Information), Deploying a Simple Web Application on Server

Unit IV

Infrastructure as Code with Terraform : Introduction to Terraform: What is Terraform, Benefits and Use Cases, Terraform Architecture and Workflow, Installing and Setting Up Terraform. Writing and Managing Terraform Configuration Files, Provisioning Infrastructure on AWS/Azure/GCP. Managing Resources and Dependencies, Terraform State and State Management. Modules and Workspaces, Terraform with Ansible for Advanced Automation. Best Practices and Security Considerations

Unit V

Introduction to Ansible : Introduction to Ansible and Configuration Management, How Ansible Works, Modern Infrastructure Management, Ansible and RedHat, Ansible Architecture, Infrastructure Management: From Shell Scripts to Ansible, Installing Ansible, Creating a Basic Inventory File, Using Ansible with Vagrant: Setup and Usage

Unit VI

Advanced Ansible : Ansible Roles and Command Line Usage, Playbooks: Structure, Writing and Running Playbooks with ansible-playbook. Real-world Playbook Examples, Handlers, Environment Variables, Variables, Facts, Prompts, Tags, Blocks, Ansible with AWS for Application Deployment

List of Practicals / Experiments:

1. Install and Configure Virtualization Platform

- Download and set up Oracle VirtualBox. Import Puppet and Ansible learning VMs

2. Puppet Installation and Initial Setup

- Install Puppet on a virtual machine and perform basic configuration tasks.

3. Develop and Apply Puppet Manifests

- Create and apply basic Puppet manifests to manage system resources.

4. Puppet Module Creation and Deployment

- Design custom Puppet modules and deploy them in a simulated environment.

5. Setup Puppet Master-Agent Architecture

- Configure Puppet Master and Agent, and test communication and certificate exchange.

6. Dynamic Resource Management Using Puppet

- Use facts and templates in Puppet to manage dynamic configurations.

7. Install and Configure Nagios Core

- Install Nagios on CentOS/Ubuntu, configure basic host and service monitoring.

8. Create Custom Nagios Plugins

- Write and integrate basic Bash or Python plugins with Nagios.

9. Monitor a Web Server with Nagios

- Configure Nagios to monitor Apache/Nginx web server status, services, and logs.

10. Use Nagios Web Interface for System Monitoring

- Manage hosts, services, comments, and view alerts through the Nagios web interface.

11. Install and Configure Ansible

- Set up Ansible, create inventory files, and use basic ad-hoc commands.

12. Write and Execute Ansible Playbooks

- Create playbooks to automate tasks like package installation and service configuration.

13. Deploy Applications Using Ansible with AWS

- Use Ansible to provision and deploy a sample web application on an AWS EC2 instance.

14. Install and Configure Terraform

- Set up Terraform and write basic configuration files to provision cloud infrastructure.

15. Use Terraform with Ansible for End-to-End Automation

- Automate infrastructure provisioning with Terraform and configure it using Ansible.

References:

1. MASTERING PUPPET 5 by RYAN RUSSELL-YATES, PACKT PUBLISHING

2. ANSIBLE FOR DEVOPS: SERVER AND CONFIGURATION MANAGEMENT FOR HUMANS by JEFF GEERLING, N.A

3. LEARNING NAGIOS by WOJCIECH KOCJAN, AMAZON.COM